

Xu Gao

Curriculum Vitae

Department of Mathematics, Shanghai Maritime University

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Experience

- 2026–Present **Lecturer**, Department of Mathematics, Shanghai Maritime University, Shanghai, China
2023–2026 **Postdoc**, School of Mathematical Sciences, Tongji University, Shanghai, China
2024 **Visiting Scholar**, Department of Mathematics, University of Pennsylvania, Philadelphia, USA

Education

- 2016–2023 **Ph.D.**, Mathematics Department, University of California, Santa Cruz, USA
Advisors: Prof. Chongying Dong & Prof. Junecue Suh
2012–2015 **M.S.**, Chern Institute of Mathematics, Nankai University, Tianjin, China
Advisor: Prof. Chengming Bai
2008–2012 **B.S.**, School of Mathematical Sciences, Nankai University, Tianjin, China

Research Interests & Selected Publications

My research lies at the intersection of algebra, geometry, and mathematical physics, with a particular focus on topics where category theory and geometry intertwine. My main results are categorized by theme as follows:

Algebraic-geometric Study of VOA Conformal Blocks

This is my primary research area. I focus on using modern algebraic geometry tools to study the geometric representation theory of Vertex Operator Algebras (VOAs) on algebraic curves and their degenerations (e.g., nodal curves, orbicurves). My current goals are to establish factorization theorems in vast generality and to develop an algebraic-geometric theory of vertex tensor categories.

- **Xu Gao**, Jianqi Liu and Yiyi Zhu, *Twisted restricted conformal blocks of vertex operator algebras I: g -twisted correlation functions and fusion rules*, **Journal of Algebra** 675 (2025), pp. 59–132.
- **Xu Gao**, Jianqi Liu and Yiyi Zhu, *Twisted restricted conformal blocks of vertex operator algebras II: twisted restricted conformal blocks on totally ramified orbicurves*, **International Mathematics Research Notices** 2026, no. 16, rnag164; arXiv:2403.00545.
- **Xu Gao**, Jianqi Liu, *A basis theorem for Genus-One Conformal Blocks and modular invariance of intertwining operators*, arXiv:2508.01294.
- **Xu Gao**, Yiyi Zhu, *Twisted and untwisted universal enveloping algebras for permutation orbifolds*, (submitted).
- **Xu Gao**, Jianqi Liu, *On strong identities of almost-canonically seminormed rings*, arXiv:2606.05965.

■ Tensor Triangular Geometry and Algebraic Topology

Applying tensor triangular geometry to study geometric invariants of stable homotopy categories in algebraic topology and representation theory.

- **Xu Gao**, Ang Li, *The stable Picard group of finite Adams Hopf algebroids with an application to the $\mathbb{R}$ -motivic Steenrod subalgebra $\mathcal{A}^{\mathbb{R}}(1)$* , **Journal of Pure and Applied Algebra** 228 (2024), no. 11, 107732.
- **Xu Gao**, *Stratification for finite groupoids*, available on webpage.

■ Bruhat-Tits Buildings and p -adic Representation Theory

This study involves combinatorial structures in non-Archimedean geometry and their applications to p -adic group representation theory. It also serves as a foundation for research in arithmetic conformal field theory.

- **Xu Gao**, *Simplicial distance in Bruhat-Tits buildings of split classical type*, **Ph.D. Dissertation**, University of California, Santa Cruz (2023).

■ Rota-Baxter Operators and Pre-Lie Algebras (Early Work)

Research on Rota-Baxter operators, pre-Lie algebra structures, and deformation theory conducted during my Master's studies.

- **Xu Gao**, Ming Liu, Chengming Bai, and Naihuan Jing, *Rota-Baxter Operators on Witt and Virasoro Algebras*, **Journal of Geometry and Physics** 108 (2016), pp. 1–20.
- **Xu Gao**, *Extensions and Non-abelian Cohomology of Pre-Lie Algebras*, **Master's Thesis**, Nankai University (2015).

■ Teaching Experience

Extensive experience in independent instruction and teaching assistance abroad. Competent in teaching courses ranging from undergraduate general education (Calculus/Linear Algebra) to advanced major courses (Number Theory/Algebra).

■ Instructor

- **University of California, Santa Cruz** Santa Cruz, USA
MATH 110: Introduction to Number Theory 2022.09 – 2023.06
Served as the sole instructor for three consecutive quarters. Responsibility includes syllabus design, lecturing, creating homework and exams, and final grading. Complete teaching materials are available on my website (gausyu.top/Teaching/).

■ Teaching Assistant

- **University of California, Santa Cruz** Santa Cruz, USA
Core Courses and Advanced Courses 2016.09 – 2022.08
Assisted in teaching core undergraduate mathematics courses and advanced courses for six years, serving over a thousand students. Courses covered include:
 - **Calculus Series**: MATH 11A (Calculus with Applications), MATH 19A/B (Calculus for Science, Engineering, and Mathematics), MATH 22 (Calculus of Several Variables);

- **Algebra Series:** MATH 21 (Linear Algebra), MATH 117 (Advanced Linear Algebra), MATH 111 (Algebra);
- **Advanced Courses:** MATH 100 (Introduction to Proof and Problem Solving), MATH 110 (Introduction to Number Theory).

Academic Activities

(Selected List)

Conference Talks

- **International Conference on Vertex Algebra, Category Theory and Related Topics**
Chern Institute of Mathematics, Nankai University, Tianjin 2026.07
Strong Identities and Algebraic Smoothing via Almost-Canonical Seminorms
- **Jingrun Star Award Ceremony and Young Scholars Forum** 2026.05
Universal Enveloping Algebras of VOAs
- **Joint Conference on Lie Theory: NKU–NYU Abu Dhabi–SUSTech** 2026.03
Universal Enveloping Algebras of VOAs
- **19th National Conference on Lie Theory** Guangdong Univ. of Tech., Guangzhou 2025.11
Orbifold Theory of VOAs via Algebro-geometric Conformal Blocks
- **AMS 2025 Summer Research Institute in Algebraic Geometry** Fort Collins, CO, USA 2025.07
Monoidal Structure on Representations of VOAs via Algebrogeometric Conformal Blocks
- **2025 International Conference on Representation Theory** Jimei University, Xiamen 2025.06
Strong Identity Condition for Almost Canonically Seminormed Rings
- **18th National Conference on Lie Theory** Tongji University, Shanghai 2023.07
 p -adic Representations and Simplicial Distance in Bruhat-Tits Buildings
- **AMS 2023 Spring Western Sectional Meeting** Fresno, CA, USA 2023.05
 p -adic Representations and Simplicial Balls in Bruhat-Tits Buildings

Invited Seminars

- **Jimei University** Xiamen 2026.05
Universal Enveloping Algebras of VOAs
- **Youth Forum on New Progress in Lie Theory** Nankai University, Tianjin 2025.12
Orbifold Theory of VOAs via Algebro-geometric Conformal Blocks
- **Seminar on Tensor Categories of VOAs** Shanghai Jiao Tong University 2025.05
Ribbon Monoidal Category via Algebrogeometric Conformal Blocks
- **Vertex Operator Algebra Seminar** Tongji University 2025.05
Ribbon Monoidal Category via Conformal Blocks and Strong Identity Condition...
- **Algebra Seminar, Chern Institute of Mathematics** Nankai University, Tianjin 2025.01
Towards a Ribbon Monoidal Category Structure via Conformal Blocks of VOAs
(Co)stratifications in representation theory
- **Math-Physics Joint Seminar, University of Pennsylvania** Philadelphia, USA 2024.10
Twisted Restricted Conformal Blocks of Vertex Operator Algebras
- **Number Theory Seminar, UC San Diego** San Diego, USA 2023.03
 p -adic Representations and Simplicial Distance in Bruhat-Tits Buildings
- **Algebra & Number Theory Seminar, University of Arizona** Tucson, USA 2023.01
 p -adic Representations and Simplicial Balls in Bruhat-Tits Buildings

Summer Schools & Other Activities

- **AMS 2024 Fall Eastern Sectional Meeting** Albany, NY, USA 2024.10
University at Albany, SUNY
- **MSRI Summer Graduate School: Sparsity of Algebraic Points** Berkeley, USA 2021.06
Mathematical Sciences Research Institute (SLMath)
- **Arizona Winter School: Topology and Arithmetic** Tucson, USA 2019.03
University of Arizona